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Quantum Logic of Word Meanings: Concept Lattices in Vector Space Models

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Abstract. This paper systematically develops the logical and algebraic possibilities inherent in vector space models for language, considerably beyond those which are customarily used in semantic applications such as information retrieval and word sense disambiguation. The cornerstone of the approach lies in a simple implementation of the connectives of quantum logic as introduced by Birkhoff and von Neumann (1936), which defines the negation of a concept as the projection onto its orthogonal subspace, and the disjunction and conjunction of two concepts as the vector sum and intersection of their subspaces. This enables us to use the full lattice structure of a vector space, bringing these models much closer to traditional semantic lattice representations such as taxonomic concept hierarchies.

We describe selected examples of this process with both negation and disjunction, and summarise experiments which show that the non-local nature of these connectives has clear advantages over their Boolean counterparts in removing the synonyms and neighbours of negated terms in information retrieval, as well as removing the negated terms themselves. Having thus validated the approach, we explore its implications for assigning semantics to some compositional phrases, showing cases where a quantum interpretation is preferable to a traditional Boolean formulation (and vice versa). Finally, we draw attention to the danger that quantum connectives may overgeneralise, and suggest another (also non-Boolean) alternative.

Keywords: Quantum Logic, Word Vectors, Non-Distributive Lattice

1. Introduction

Vector spaces have become widespread in natural language processing: initially used to assign coordinates to queries and documents for information retrieval (Salton and McGill, 1983), variants of the vector model have become used for modelling human language acquisition (Landauer and Dumais, 1997), word sense disambiguation (Schütze, 1998) and for the automatic mapping of unknown words into a concept hierarchy (Widdows, 2003c). These models have only made use of a small part of the algebraic structure inherent in vector spaces. The only algebraic operations which are standardly performed on word vectors have been commutative addition (for combining several word vectors into a ‘document vector’), and taking the scalar product of two vectors to measure



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their similarity (for matching query vectors to document vectors or for generating a ‘similarity thesaurus’). There have been some attempts to represent logical connectives more faithfully than is possible using just commutative addition: in particular, Salton et al. (1983) used p -norms of varying exponents to distinguish between conjunctions and disjunctions, and Dunlop (1997) considered the effect of subtracting a constant multiple of a vector to represent its negation, though neither of these works presents a full logic for vector spaces.

However, if we regard a vector space as a collection of nested subspaces of different dimensions rather than as a set of points,¹ vector spaces have a simple lattice structure, the *join* $A \vee B$ of two subspaces being their vector sum $A + B$ and their *meet* $A \wedge B$ being the standard intersection $A \cap B$. Once a vector space is equipped with a scalar product, this can be used to define the negation $\neg A$ of a subspace as its orthogonal complement $A^\perp$, and these three operators define a logic on the subspaces of V . Birkhoff and von Neumann (1936) described this structure, giving it the name *quantum logic* because (in the eyes of some scholars) it can be used to model and explain some of the characteristics of quantum mechanics (Putnam, 1976). The most important features of quantum logic, and its differences with classical Boolean logic, are discussed in Section 2.

In Section 3 we review methods for building vector space models from a corpus or document collection, focussing on the WORDSPACE developed by Schütze (1998) which uses latent semantic analysis (Landauer and Dumais, 1997). The WORDSPACE model represents each word as a linear combination of ‘latent semantic axes’, much as the *state* of a particle in quantum mechanics is represented by a linear combination of ‘pure states’. The ‘semantic similarity’ between two word vectors can be measured automatically using the cosine of the angle between the vectors.

In Section 4, we explain how to encode the quantum connectives as operations on vectors in this WORDSPACE. In particular, we explain how negation and (negated) disjunction can be implemented using only elementary linear algebra and the WORDSPACE we have already built. The linguistic effect of these operations can be investigated immediately, and in Section 5 examples are presented in which a search engine equipped with quantum negation is used to remove predominant senses of ambiguous words, enabling a user to obtain word vectors for rarer senses. This technique is extended using quantum disjunction and negation together to remove *several* vectors, enabling a user to

¹ In terms of the historical foundations of the theory of vectors, this follows the tradition of Grassmann’s *Ausdehnungslehre* more closely than that of Hamilton’s *quaternions* (Smith, 1959, 677-696).

prune away unwanted features and gradually ‘home in’ on their desired meaning. Encouraged by these successes, large scale document retrieval experiments were set up which demonstrate that the while Boolean negation (trivially) removes unwanted terms from a pool of retrieved documents, quantum negation does much better at removing the synonyms and neighbours of the unwanted terms as well those terms themselves, which is strong evidence that quantum negation is more successfully computing with meaning.

Finally in Section 6 we consider some of the situations where Boolean and quantum logic give different interpretations for the meaning of compositional phrases, concluding that quantum logic is a more suitable model for natural language in at least some of these cases.

2. Quantum logic in vector spaces

In this section we introduce the connectives of quantum logic, which are particularly appropriate for use within vector spaces. Quantum mechanics is, formally speaking, a theory of *states* and operators upon states in vector spaces. A state is represented by a vector whose coordinates are complex numbers, rather than the Boolean values of classical logic. For example, whereas a classical ‘bit’ or ‘binary digit’ can be in only one of two states represented by the values 0 and 1, the state of a *quantum bit* or *qubit* can take the more general form

$$\alpha_0|0\rangle + \alpha_1|1\rangle \tag{1}$$

where $|\alpha_0|^2 + |\alpha_1|^2 = 1$ and $|0\rangle$ and $|1\rangle$ are orthonormal basis vectors (Hirvensalo, 2001, p. 8). The vast majority of vector spaces which have been used to model linguistic data use coordinates in the real numbers $\mathbb{R}$ rather than the more general complex numbers $\mathbb{C}$, so for the sake of mathematical simplicity we will work with real numbers throughout this paper.²

An observation of the qubit will find the particle to be in one of the basis states $|0\rangle$ and $|1\rangle$, the probability of each outcome being given by the real numbers $|\alpha_0|^2$ and $|\alpha_1|^2$ respectively. In the collapse from a continuous set of possible states to a discrete set of possible empirical observations, the state becomes *quantised*. The basis states $|0\rangle$ and $|1\rangle$

² Another mathematical distinction is that quantum mechanics is usually modelled within a *Hilbert space*, which is a complete inner product space (Cohen, 1989, 2.18). All the vector spaces we will use in this paper are Hilbert spaces, using the standard Euclidean scalar product as an inner product to measure similarity, so for simplicity we will continue to use the term ‘vector space’ rather than ‘Hilbert space’.

are called *pure states*: conversely, all other states (those in which both α_0 and α_1 are non-zero) are called *mixed states*.

The collection of all possible mixed states consistent with Equation 1 is the *quantum disjunction* of the states $|0\rangle$ and $|1\rangle$. Note that the quantum disjunction is *more general* than the classical disjunction: whereas the classical disjunction consists only of the pure states $|0\rangle$ and $|1\rangle$, the quantum disjunction allows for linear combinations of these pure states.³ In other words, mixed states are *included* by the quantum disjunction but *excluded* by the classical disjunction. This difference is encapsulated formally by the *distributive law* (Partee et al., 1993, p 18):

$$C \wedge (A \vee B) \equiv (C \wedge A) \vee (C \wedge B), \quad (2)$$

which classical Boolean logic obeys but quantum logic does not.

Quantum logic was formally introduced by Birkhoff and von Neumann (1936) as a framework in which to account for the observations and predictions of quantum mechanics, which exhibits some distinctly non-classical behaviour which can be accounted for if the distributive law is violated. A famous example is given by the *two-slit experiment*, in which a single particle e has two possible paths A and B available to it. The outcomes ‘ e passed through A ’ and ‘ e passed through B ’ are represented by the pure states $|A\rangle$ and $|B\rangle$. However, interference patterns are observed which can only be accounted for if the particle e is in a mixed state $\lambda|A\rangle + \mu|B\rangle$, resulting in the conclusion that ‘part of e passed through A and part of e passed through B ’. This violates the distributive law, so the quantum disjunction of $|A\rangle$ and $|B\rangle$ gives a more appropriate model for this experiment than their classical disjunction, as argued by Putnam (1976).

The equivalence between Boolean logic and set theory (guaranteed by the Stone Representation Theorem (Partee et al., 1993, Ch 12)) can be used to account for many of the properties of classical logic. In a similar fashion, once we accept the validity of the vector space model, the structure of quantum logic itself is quite simple and is arrived at precisely by replacing the notions of sets and subsets with those of vector spaces and subspaces. (The mathematics of this process is described thoroughly by Birkhoff and von Neumann (1936) and Cohen

³ A simple way to envisage the difference between these two forms of disjunction is to consider the possible trajectories of a point which starts in the centre of a map with the instructions that it can travel in a North-South *or* an East-West direction. If this disjunction is interpreted classically, the particle can only travel to one of a ‘cross-shape’ of points which are either of the same latitude or of the same longitude as the starting point. In the disjunction is interpreted in the quantum framework, the point can travel anywhere that is a linear combination of north-south or east-west journeys, *i.e.* anywhere on the map.

(1989): it is approached from a more philosophical point of view by Putnam (1976) and Wilce (2003).) Events in quantum mechanics are represented by subspaces of a vector space V . This leads us to consider the set $L(V)$ of subspaces of vector space V , which is a partially ordered set under the inclusion relation, so that an event A implies an event B precisely when $A \leq B$.

The set $L(V)$ is naturally a *lattice*, in the sense of (Birkhoff, 1967; Davey and Priestley, 1990). The greatest lower bound or *meet* of $A, B \in L(V)$ is the greatest element $C \in L(V)$ such that $C \leq A$ and $C \leq B$, which is precisely the intersection $A \cap B$. The least upper bound or *join* of A and B is the smallest $D \in L(V)$ such that $A \leq D$ and $B \leq D$. However, the set union $A \cup B$ is not in general a member of $L(V)$, and the smallest member of $L(V)$ which contains this set is instead the linear span $A + B$. Lattices are widely used as models for linguistic information, with examples including taxonomies or concept hierarchies (such as WordNet (Fellbaum, 1998)) and the lattices used in Formal Concept Analysis (Ganter and Wille, 1999). This paper draws attention to the fact that vector spaces such as WORDSPACE can be regarded as examples of this general semantic model.

If the vector space V is also equipped with an inner product, for each $A \in L(V)$ we can define its (unique) orthogonal complement (Jänich, 1994, §8.2) as its logical complement or *negation*. This gives three connectives on the lattice $L(V)$, defined as follows (Birkhoff and von Neumann, 1936, §1, §6) (Putnam, 1976, p. 178):

$$\begin{array}{ll} \text{Conjunction} & A \wedge B = A \cap B \\ \text{Disjunction} & A \vee B = A + B \\ \text{Negation} & \neg A = A^\perp \end{array} \quad (3)$$

It is simple to show that these connectives on $L(V)$ satisfy the necessary axioms (such as, for example, $A + A^\perp = V$, $A \cap A^\perp = \{0 \in V\}$) for the lattice $L(V)$ to be a logic on $L(V)$ (Cohen, 1989, p. 36).

An important equivalence is given by identifying $A \in L(V)$ with the unique orthogonal projection map $P_A : V \rightarrow A$, and through this bijection the logic of subspaces $L(V)$ gives rise to the logic of projection mappings on V . This logic plays a key role in quantum mechanics, where the image $P_A(V)$ of a point $v \in V$ under the projection P_A is used to measure the probability that a particle in the state represented by v will be found to have a physical property represented by the subspace A , using the scalar product $P_A(v) \cdot v$ as a probability measure (Wilce, 2003). The *qubit* described by Equation 1 gives an example of such a measure.

One important property which therefore holds in classical (Boolean) logic but not in quantum logic is the *commutative law* (Partee et al.,

1993, p. 18). The commutative property does not apply to quantum logic because two projection mappings P_A and P_B do not in general commute. (An easy example is to consider the projections onto the lines $x = 0$ and $y = x$ in the plane $\mathbb{R}^2$.) This is used to account for the fact that observations interfere with one another in quantum mechanics, which leads to Heisenberg's famous uncertainty principle (Birkhoff and von Neumann, 1936, §1). Measurements made by the projections P_A and P_B are said to be *compatible* if and only if P_A and P_B commute, which imposes particular conditions on the subspaces A and B (Cohen, 1989, p. 37).

The differences between the negation and disjunction operators of classical and quantum logic can be (more or less adequately) described in terms of similarity. In set theory, $\neg A$ corresponds simply to those things which are not elements of A , given by the complement A' . In quantum logic, $\neg A$ is the narrower concept of things which have *nothing in common* with the elements of A . Similarly, in classical logic, the disjunction $A \vee B$ contains only those things that are either members of A or members of B , whereas the quantum disjunction $A \vee B$ also includes those things which share the features of P and Q . The quantum negation and disjunction connectives therefore have an effect on points outside the sets which are being negated or disjoined: this wider influence can be used to account for the *non-locality* of quantum theory.

3. Representing word meaning in vector spaces

In this section we review the techniques for building the word vector space or 'WORDSPACE' in which we investigate the behaviour of word meanings under the action of the quantum connectives. The first linguistic examples of vector spaces were developed for information retrieval (Salton and McGill, 1983; Baeza-Yates and Ribiero-Neto, 1999). Counting the number of times each word occurs in each document gives a *term-document matrix*, where the i, j^{th} matrix entry records the number of times the word w_i occurs in the document d_j . The rows of this matrix can then be thought of as *word vectors*. The dimension of this vector space (the number of co-ordinates given to each word) is therefore equal to the number of documents in the collection. *Document vectors* are generated by computing a (weighted) sum of the word vectors of the words appearing in a given document.

Typically, such term-document matrices are extremely sparse. The information can be concentrated in a smaller number of dimensions using (among other dimension-reduction algorithms) singular value decomposition (Trefethen and Bau, 1997, p. 25), projecting each word

onto the n -dimensional subspace which gives the best least-squares approximation to the original data. This represents each word using the n most significant ‘latent variables’, and for this reason this process is called *latent semantic analysis* (Landauer and Dumais, 1997). More information about the vector space model generally and latent semantic analysis in particular can be found in (Manning and Schütze, 1999, Ch 15).

These techniques are used in information retrieval to measure the similarity between words (or more general query statements) and documents, using a similarity measure such as the cosine of the angle between two vectors

$$\text{sim}(q, d) = \frac{\sum q_i d_i}{\sqrt{\sum q_i^2 \sum d_i^2}} = \frac{q \cdot d}{\|q\| \|d\|},$$

where q_i, d_i are the co-ordinates of the vectors q and d , and $q \cdot d$ is the (Euclidean) scalar product of q and d and $\|q\|$ is the norm of the vector q . This calculation is simplified further by normalising all vectors to have unit length, so that the cosine similarity is the same as the scalar product. This is a standard technique which (for example) avoids giving too much semantic significance to frequent terms or long documents. Normalised vectors were used in all of the models and experiments described in this paper. Note that this coincides exactly with the standard procedure for representing quantum states (as in Equation 1).

A natural advantage of this structure is that it can be used to define a similarity score between pairs of terms in exactly the same way — two terms will have a high similarity score if they often occur in the same documents, and only seldom occur without one another. In general, several terms are combined into a combined query-statement using commutative vector addition (though the fuzzy-set and p -norm operations of (Salton et al., 1983) give more sophisticated models for conjunction and disjunction which also combine some of the benefits of Boolean and vector approaches).

A variant of the traditional term-document matrix was developed by Schütze (1998) specifically for the purpose of measuring semantic similarity between words. Instead of using the documents as column labels for the matrix, semantically significant *content-bearing words* are used, and other words in the vocabulary are given a score each time they occur within a context window of (say) 15 words of one of these content-bearing words. Thus the vector of the word *football* is determined by the fact that it frequently appears near the words *sport* and *play*, etc. Such a vector space where points are used to represent words and concepts will be called a WORDSPACE. The examples and experiments described

in this article use exactly this sort of WORDSPACE, using the cosine measure to compute similarity.

This WORDSPACE was used by Schütze (1998) to perform word sense discrimination, as follows. The ‘senses’ of an ambiguous word w are identified by clustering the vectors composed of sentences containing w . In this way, a discrete set of possible senses for w are discovered and represented by the centroids of their corresponding cluster. Previously unseen instances of w are then disambiguated by mapping them to the nearest centroid according to the cosine measure. This corresponds exactly with the quantum measure $P_A(v) \cdot v$ which gives the probability that an event v will be in the state given by A , since the cosine measure gives the length of the projection of v onto A . (This also raises the possibility that Schütze’s ‘sense vectors’ could be meaningfully generalised to ‘sense subspaces’ if these could also be learned from text.)

One of the great strengths of this method is that the cosine measure returns a *continuous* relevance score which in information retrieval allows documents to be ranked according to how relevant they are, rather than simply partitioned into ‘relevant’ (matching) and ‘non-relevant’ (non-matching) documents, an early technique which is not scalable to huge document collections like the internet. This generalisation from Boolean values in the set $\{0,1\}$ to continuous values in the interval $[0,1]$ is one of the hallmarks of modern retrieval systems, and also of quantum mechanics, in which the guaranteed prediction of an outcome in classical physics is replaced by a probability that an outcome will be observed. In this way, both information retrieval and quantum mechanics benefit from the vector space model as a way to measure and compare distances between events, which allows the influence of an event to be extrapolated from the point set on which that event is defined to the whole space, thus allowing for non-local effects and mixed states.

4. The quantum connectives in wordspace

In this section we form equations which describe the behaviour of the quantum logical connectives in WORDSPACE, giving explicit details for calculating negation and disjunction.

4.0.1. *Vector Negation*

Suppose we want to model the meaning of a statement like *rock NOT music* in such a way that the system realises we are interested in the geological, not the musical meaning of the word *rock*. This is accomplished by modelling the ‘mixed state’ of an ambiguous word like *rock* as

a linear sum of ‘pure states’ corresponding to the musical and geological meanings of *rock*. If we assume that these two components of the *rock* vector are orthogonal to one another, we can then assign a unique vector to the statement *rock NOT music* by projecting the vector for *rock* onto the subspace orthogonal to the vector for *music*. This process is described in detail in the following paragraphs. We begin by pointing out the correspondence between notions of ‘irrelevant’ and ‘orthogonal’ in WORDSPACE.

DEFINITION 1. *Two words a and b are considered irrelevant to one another if their vectors are orthogonal, i.e. a and b are mutually irrelevant if $a \cdot b = 0$.*

Using this definition, two words are irrelevant to one another if they have no features in common at all, just as a document is regarded as completely irrelevant to a user if its cosine similarity with the user’s query is zero — precisely when the query vector and the document vector are orthogonal. The statement “ $A \text{ NOT } B$ ” (written as ‘ $a \wedge (\neg b)$ ’⁴) can now be interpreted as “those features of a to which b is irrelevant”.⁵

DEFINITION 2.

Let V be a vector space equipped with a scalar product. For a vector subspace $A \subseteq V$, define the orthogonal subspace $A^\perp$ to be the subspace

$$A^\perp \equiv \{v \in V : \forall a \in A, a \cdot v = 0\}.$$

Let A and B be subspaces of V . By $\neg B$ we mean $B^\perp$ and by $A \wedge \neg B$ we mean the projection of A onto $B^\perp$.

Let $a, b \in V$. By $a \wedge \neg b$ we mean the projection of a onto $\langle b \rangle^\perp$, where $\langle b \rangle$ is the subspace $\{\lambda b : \lambda \in \mathbb{R}\}$.

We now show how to use these notions to perform calculations with individual vectors in WORDSPACE, using a standard projection mapping technique (Jänich, 1994, §8.2).

THEOREM 1. *Let $a, b \in V$. Then $a \wedge \neg b$ is represented by the vector*

$$a \wedge \neg b \equiv a - \frac{a \cdot b}{|b|^2} b.$$

where $|b|^2 = b \cdot b$ is the modulus of b .

⁴ For syntactic reasons it is preferable to write $a \wedge (\neg b)$ (corresponding to the query statement “ A and not B ” or “ A but not B ”) instead of just $a \neg b$ (“ A not B ”).

⁵ This idea of negation as ‘otherness’ is found in Plato’s *Sophist* dialogue (Horn, 2001, p. 1).

Proof. Taking scalar product with b , we have that

$$\begin{aligned} (a \wedge \neg b) \cdot b &= \left(a - \frac{a \cdot b}{|b|^2} b\right) \cdot b \\ &= a \cdot b - \frac{(a \cdot b)(b \cdot b)}{b \cdot b} \\ &= 0. \end{aligned}$$

This shows that $a \wedge \neg b$ and b are orthogonal, so the vector $a \wedge \neg b$ is precisely the part of a which is irrelevant to b (in the sense of Definition 1) as desired.

For normalised vectors, Theorem 1 takes the particularly simple form

$$a \wedge \neg b = a - (a \cdot b)b.$$

In practice this vector is then renormalised for consistency. As well as being well-motivated theoretically, this expression for negation is computationally extremely efficient to implement in the ‘search phase’ of a retrieval system. In order to find concepts, terms or documents that are closely related to $a \wedge \neg b$, it is not necessary to compare each candidate with both a and b and then compute some difference. Theorem 1 gives a single vector for $a \wedge \neg b$, so finding the similarity between any other vector and $a \wedge \neg b$ is just a single scalar product operation.

4.0.2. Vector Disjunction

Modelling disjunctive expressions works similarly. Disjunction in Boolean logic is modelled as the union of sets, which corresponds in quantum logic to the vector sum of subspaces.

DEFINITION 3. *Let A and B be subspaces of V . The disjunction $A \vee B$ is represented by the subspace*

$$A + B = \{a + b : a \in A, b \in B\}.$$

Let $b_1 \dots b_n \in V$. The disjunction $b_1 \vee \dots \vee b_n$ is represented by the subspace

$$B = \{\lambda_1 b_1 + \dots + \lambda_n b_n : \lambda_i \in \mathbb{R}\}.$$

Computing the similarity between a vector a and this subspace B is not entirely straightforward — the simple solution of using cosine similarity to measure the projection of a onto the span of each of the

basis vectors b_i in turn and adding the results is only correct if the set $\{b_i\}$ form an orthonormal basis for B .⁶

One successful approach to this problem has been to first use the Gram-Schmidt process (Jänich, 1994, p. 142) to obtain an orthonormal basis $\{\tilde{b}_i\}$ for the subspace B . Computing the similarity between a vector a and a subspace B is still computationally more expensive than for negation, because the scalar product of a with $n = \dim B$ vectors must be computed. Thus the gain we get by comparing each document with the query $a \wedge \neg b$ using only one scalar product operation is lost for disjunction, though in Section 5 we show that this desirable property is recovered for *negated* disjunction.

4.0.3. Vector Conjunction

We make a few remarks about vector conjunction, though in practice our experiments in WORDSPACE have so far focussed only on negation and disjunction. Just as disjunction makes things more general, we would expect conjunction to make them more specific. Since our underlying WORDSPACE is homogeneous (in the sense that any two non-zero points can be mapped to each other by a linear transformation), no one point is naturally any more or less general than any other. This is one of the noticeable drawbacks for the basic vector model: the terms *plant*, *fruit* and *apple* are all represented by single points without any notion of inclusion or inheritance. Ideally, this problem could be solved by having concepts represented not only by points but also by higher dimensional subspaces. Then *plant*, for example, could refer to a space with *fruit* as a subspace thereof, and with *apple* as an even smaller subspace or point in the *fruit* subspace. In theory it should be possible to build such a space using a taxonomy and corpus-data, though to our knowledge this has not been accomplished. Such a structure would present a natural model for conjunction: the conjunction of two subspaces B_1 and B_2 would simply be their intersection $B_1 \cap B_2$, and so the similarity between a point a and the subspace $B_1 \cap B_2$ would be given by the cosine measure $P_{B_1 \cap B_2}(a) \cdot a$.

Unless B_1 and B_2 are orthogonal, the projection $P_{B_1 \cap B_2}(a)$ will be in general be different from the composition $P_{B_2}(P_{B_1}(a))$. This latter formulation may also be a worthwhile model for conjunction, which naturally allows for both non-commutativity and negational descriptors. These possibilities are described in Section 6.

⁶ This is another feature of the non-commutativity of quantum logic. Observables which *can* be measured separately in this fashion are called *compatible* (Cohen, 1989, p. 37).

5. Experiments and Evaluation

This section presents initial examples of our theory which demonstrate the effectiveness of vector negation, and of vector disjunction and negation together, at finding vectors which represent different senses of ambiguous words.

WORDSPACE models were built as described in Section 3 using the New York Times (NYT) newswire from the Linguistic Data Consortium, a corpus consisting of some 173 million words from news articles written between July 1994 and December 1996, and the British National Corpus (BNC) available from Oxford University, consisting of some 100 million words drawn from a wide variety of British English. As one might expect, news articles consistently prefer some meanings of ambiguous words over others: for example, the word *suit* in the NYT corpus is used far more often in a legal context than a clothing context. To test the effectiveness of our negation operator, we tried to find some of the less common meanings by removing words belonging to the more predominant meanings.

Table I. Terms related to *suit NOT lawsuit* (NYT) and to *plant NOT chernobyl* (BNC)

suit			suit NOT lawsuit		plant			plant NOT chernobyl	
suit	1.000000		pants	0.810573	plant	1.000000		plant	0.821560
lawsuit	0.868791		shirt	0.807780	plants	0.823916		plants	0.721080
suits	0.807498		jacket	0.795674	chernobyl	0.570122		vegetation	0.544346
plaintiff	0.717156		silk	0.781623	reactor	0.538019		flowering	0.501036
sued	0.706158		dress	0.778841	reactors	0.535419		planting	0.499217
plaintiffs	0.697506		trousers	0.771312	planting	0.513893		abundant	0.498105
suing	0.674661		sweater	0.765677	generating	0.512973		cultivation	0.492297
lawsuits	0.664649		wearing	0.764283	radioactive	0.502536		habitat	0.485266
damages	0.660513		satin	0.761530	nuclear	0.502159		leaves	0.480611
filed	0.655072		plaid	0.755880	seafield	0.500940		machinery	0.478653
behalf	0.650374		lace	0.755510	machinery	0.500783		temperate	0.477524

Table I shows that vector negation is very effective for removing the ‘legal’ meaning from the word *suit* and the ‘manufacturing installation’ meaning from the word *plant*, leaving respectively the ‘clothing’ and ‘vegetation’ meanings. Note that negating a particular word also removes concepts related to the negated word. This gives credence to the claim that our mathematical model is removing the *meaning* of a word, rather than just a string of characters, using non-locality to propagate the effect of removing an individual term throughout the WORDSPACE.

Vector negation and disjunction can be combined to remove several unwanted areas of meaning simultaneously. Suppose we negate not only one argument but several. If a user specifies that they want documents related to *a* but not $b_1, b_2, \dots, b_n$, then (unless otherwise stated) it is clear that they only want documents related to *none* of the unwanted terms b_i (rather than, say, the average of these terms). This is enabled by the de Morgan equivalence (Partee et al., 1993, p. 18) which holds

Table II. Terms related to *rock*, *rock NOT band* and *rock NOT band, arkansas* in the NYT

rock		rock NOT band		rock NOT band, arkansas	
rock	1.000000	rock	0.450473	rock	0.412383
band	0.892790	dubious	0.402324	stands	0.389242
band's	0.868856	arkansas	0.400669	celestial	0.387825
bands	0.867765	ark	0.392304	underground	0.381206
punk	0.861354	madison	0.378165	muck	0.376508
pop	0.848222	celestial	0.376519	touches	0.373402
guitar	0.840769	muck	0.367648	pure	0.373129
tunes	0.837099	sheds	0.363119	wind	0.373017
reggae	0.828602	whitewater	0.362743	echoes	0.360734
acoustic	0.820719	gore	0.360440	explosions	0.356637
blues	0.817073	wind	0.357299	beneath	0.355244
rockers	0.807684	majestic	0.355958	planet	0.354783

Not surprisingly, the word *rock* is most closely associated with pop music in the New York Times corpus. However, removing these meanings by negating the word *band* leaves a set of associations derived from the town *Little Rock, Arkansas*. (The word *little* is on the stop-list of frequent, non meaning bearing words and so does not appear (Baeza-Yates and Ribiero-Neto, 1999, p. 167).) Removing *arkansas* as well gives meanings closely associated to rock as a geological material.

in Boolean and quantum logic, whereby the expression

$$a \wedge (\neg b_1) \wedge (\neg b_2) \wedge \dots \wedge (\neg b_n)$$

is equivalent to

$$a \wedge (\neg(b_1 \vee \dots \vee b_n)). \quad (4)$$

Using Definition 3 to model the disjunction $b_1 \vee \dots \vee b_n$ as the vector subspace $B = \{\lambda_1 b_1 + \dots + \lambda_n b_n : \lambda_i \in \mathbb{R}\}$, this expression can be assigned a unique vector which is orthogonal to *all* of the unwanted arguments $\{b_j\}$. This technique can be used to ‘home in on’ the desired meaning by systematically pruning away unwanted features (see Table II).

5.1. EXPERIMENTS WITH DOCUMENT RETRIEVAL

Encouraged by these successes on a few test-words, we performed large-scale experiments to investigate the potential of vector negation and disjunction for removing unwanted concepts from retrieved documents. The results are reported in much more detail in a separate paper (Widdows, 2003b). In order to evaluate the effectiveness of different forms of negation, we used the hypothesis that a query for

keyword a NOT keyword b

should retrieve documents containing many examples of keyword *a* and few examples of keyword *b*. This can be accomplished trivially by removing any documents that contain keyword *b* after retrieval, in the traditional Boolean manner (Salton and McGill, 1983, p. 26).

However, we also measured the occurrence of *synonyms* and *neighbours* of the term *b*. Vector negation removed dramatically more of these than the Boolean method (giving an improvement of up to 76% for queries with multiple negated arguments), which we believe to be strong evidence that vector negation as described in this paper removes not only unwanted terms but unwanted *areas of meaning*.

6. Analysis and Conclusions

The examples in Section 5 demonstrate that the quantum connectives can manipulate word meanings to obtain different senses of word which correspond closely with our intuitions, and the retrieval experiments outlined in Section 5.1 demonstrate that this is not a feature of just a few chosen words but that the quantum connectives in WORDSPACE are successful at a large-scale document retrieval task. It therefore makes sense to investigate other linguistic situations where the quantum connectives might be of use in modelling the semantics of compositional phrases, and we conclude this paper by examining some of the possible situations which give theoretical support — or point out some of the flaws — with using quantum logic as an alternative to classical logic for semantic modelling.

As models for composition of meaning, Boolean and quantum connectives seem to have different spheres of influence. One intuitive prediction, based on the mathematical models underlying the two frameworks, is that Boolean connectives should be more appropriate for describing discrete entities, and quantum connectives should describe concepts which are more continuous. This prediction is borne out in at least some examples. If one's host for a dinner party said "*Please come at 7 or 7.30*", one would expect 7.15 also to be a perfectly agreeable time to arrive, which would be false under a Boolean interpretation of "*7 or 7.30*" but true under a quantum interpretation. On the other hand, if upon arrival one was asked "*Would you like an apple or a plum?*" and responded positively, one would not really expect to be given a *nectarine* on the basis that *nectarines* are contained in the lattice disjunction *apples* $\vee$ *plums* = *fruit*. In this situation we are talking about discrete objects and it appears that a Boolean interpretation is appropriate, though it is also reasonable to claim that a 'universal taxonomy' in which to form disjunctions such as *apples* $\vee$ *plums* = *fruit*

is not an appropriate model for a context where there are only a few specific fruits available. These two explanations are related — in a local context or domain some sets of objects may become ‘well-formed subspaces’ in that particular context, adapting a lattice to give more restricted disjunctions which are more Boolean than quantum in nature. In practice, there are many other factors to take into account — for example, in many day to day situations (such as train timetables), a continuous variable becomes ‘quantised’ and interpretations change accordingly.

We have drawn some attention to the non-commutativity of quantum logic. It well-known that commutative models of composition are inadequate for language — a *Venetian blind* is not a *blind Venetian*, a *horse chestnut* is not a *chestnut horse*, etc. Both the sum of two vectors and the Boolean intersection of two sets are commutative operations, and thus the situations in which they give an accurate model for composition are severely limited (see (Cruse, 2000, §4.2)). However, methods for composition of vectors are by no means limited to the standard vector sum operation (more sophisticated product operations which are still used are given in Grassmann’s original work on *Ausdehnungslehre* (Smith, 1959, 684-696)). One standard way of combining two vectors is to take their *tensor product*, whose role in quantum mechanics it outlined by (Hirvensalo, 2001, p. 10,11). The tensor product of two vectors has also been used as a model for distributed representation in cognitive science (Plate, 2003) and in a specifically quantum description of conjunction of concepts (Gabora and Aerts, 2002). As well as being non-commutative, the tensor product allows for *entanglement*, whereby new quantum states can occur in the tensor product $A \otimes B$ with properties which are not strictly inherited from either A or B , but from their combination. Gabora and Aerts (2002) propose this as a possible solution to the “*guppy problem*”: to give an explanation of how a *guppy* can be regarded as a prototypical *pet fish* even though it is neither a prototypical *pet* nor a prototypical *fish*.

A related problem for the Boolean framework is that of *negational descriptors*, which occur with phrases whose meaning is semantically compositional but cannot be modelled by simple set intersection (for example, a *fake Ming vase* is not really a *Ming vase* and a *stone lion* is not really a *lion*, it is just shaped like one). This problem can be addressed using the non-locality inherent in the quantum framework. For example, even though the set representing *stone lion* should not be a subset of the set representing *lion*, the projection from the *lion* set onto the *stone lion* set may still be non-zero and give useful information about which *features* of *lion* are compatible with the modifier *stone*, allowing

for a correct compositional interpretation. This solution is suggested in (Gärdenfors, 2000, §4.4).

There are remaining conceptual drawbacks with the quantum connectives which should be addressed. For example, the assumption in Section 4.0.1 that the different meanings of an ambiguous word are orthogonal to one another may be too strong in many cases, particularly when the ambiguity is systematic rather than accidental (Pustejovsky, 1995, Ch 3).

One danger with the quantum disjunction operator is that in a WORDSPACE of n -dimensions, n fairly similar concepts could be used to generate the whole space, provided they are linearly independent. Since the statistical chances of a collection of vectors learnt from corpora being exactly linearly dependent, we are left with the possibility that the ‘disjunctions’ predicted by quantum logic may become far too general. For example, the standard vector formalism gives only a two dimensional subspace as the disjunction of three lines which were exactly coplanar, but gives a whole extra dimension for the disjunction of three lines which are perturbed however slightly from being coplanar. Such chaotic behaviour is probably undesirable, and at some point it may be worth ignoring word-vectors which are ‘nearly linearly dependent’ on previous ones as being redundant.

Another problem with the ‘linear span of the arguments’ approach to disjunction is that it permits interpolation *and* extrapolation, where extrapolation may be inappropriate. For example, in the “7 or 7.30” example, we should not predict that 6 o’clock is also an acceptable arrival time. It follows that a better option might be to interpret such a disjunction not as a linear subspace but as a *simplex* by adding the conditions $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$ to Definition 3, which ensures that the disjuncts are a boundary as well as a generating set for their disjunction.

There are many apparent similarities between the historical debate over quantum and classical mechanics on the one hand, and the tension between ‘symbolic’ and ‘statistical’ approaches to natural language processing on the other. The vector model for information retrieval was first adopted largely because it allowed for a naturally continuous “relevance score” rather than a simple dichotomy between relevance and irrelevance, in much the same way that quantum mechanics yields a probability that a particular event will be observed. The analogy between finding the ‘state’ of a particle through measurement, and finding the ‘sense’ of an ambiguous word in context, has received attention from both Gabora and Aerts (2002) and Widdows (2003a), and as we have pointed out, the disambiguation methods of Schütze (1998) use exactly the same equations as those used to give the probability of a particular event in quantum mechanics. More generally, quantum

mechanics is possibly the single most successful scientific theory for making rigorous, testable predictions about systems where it is known that *exceptions are always a possibility*. That natural language bears the hallmarks of such a system is at the least plausible.

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